MAHARASHTRA STATE BOARD OF TECHNICAL EDUCATION, MUMBAI

A Project Report on

Event Management System

Submitted in partial fulfillment for the award of

Diploma in Computer Engineering

By

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Kalpesh Satish Patil

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DEPARTMENT OF COMPUTER ENGINEERING

S.S.V.P.S's Bapusaheb Shivajirao Deore Polytechnic, Dhule

(Institute Code : 0059)

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I would like to express my sincere gratitude to everyone who supported me throughout the development of my capstone project titled “Event Management System with Geofencing.” I am especially thankful to my project guide for their valuable guidance, continuous encouragement, and support during the entire project work. Their suggestions and insights helped me understand the concepts clearly and improve the quality of my project. I also extend my thanks to all the faculty members for providing the necessary knowledge and resources required for completing this project successfully. I am grateful to my friends and classmates for their help, ideas, and technical support whenever needed. I would also like to thank my family for their constant motivation, patience, and encouragement throughout the project duration. This project has helped me gain practical knowledge and enhance my technical skills in real-world applications. Finally, I would like to thank everyone who directly or indirectly contributed to the successful completion of this project.

Anup Sanjay Rajput

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Abstract

The “Event Management System with Geofencing” is a web-based application designed to simplify and automate the process of managing events, participant registrations, and attendance tracking. Traditional event management systems often rely on manual attendance or basic QR scanning methods, which are time-consuming and prone to errors and misuse. This system introduces a secure and efficient solution by integrating geofencing technology, ensuring that participants can mark their attendance only when they are physically present within the defined event location.

The system provides features such as event creation, user registration, role-based access for admin and organizers, QR-based attendance, and real-time location validation. All data is stored in a centralized database, allowing easy access, management, and retrieval of information. The use of geofencing enhances security by preventing fake attendance and improving the reliability of event records.

Additionally, the system improves coordination between organizers and participants, reduces manual effort, and enhances overall efficiency. It is developed using modern web technologies, making it scalable, user-friendly, and suitable for various types of events such as college programs, workshops, and seminars. Overall, this system provides a smart and reliable solution for modern event management.

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Chapter 1

Introduction

Event management has become an essential part of modern organizations, especially in educational institutions, corporate sectors, and social communities. Managing events efficiently requires proper handling of participant registration, scheduling, communication, and attendance tracking. Traditional methods of event management often rely on manual processes, which are time-consuming, error-prone, and difficult to manage as the number of participants increases.

In many cases, attendance is recorded manually or through simple systems that do not verify the actual presence of participants. This creates issues such as proxy attendance, inaccurate records, and lack of transparency. As events grow in scale, managing such data becomes increasingly complex and inefficient, leading to poor coordination between organizers and participants.

To overcome these challenges, the Event Management System with Geofencing is introduced as a modern and efficient solution. This system automates the entire event management process, from event creation to attendance tracking, using a digital platform. It eliminates the need for manual record-keeping and ensures that all data is stored securely in a centralized database.

One of the key features of this system is geofencing, which uses GPS technology to define a virtual boundary around the event location. Participants can mark their attendance only when they are physically present within this boundary. This ensures authenticity and prevents fake attendance, making the system more reliable and secure.

The system also provides role-based access for different users such as admin, organizer, and participants. Admin can manage the overall system, organizers can create and manage events, and participants can register and attend events بسهولة. This structured approach improves coordination and makes the system user-friendly.

Chapter 2

Literature Survey

2.1 Study of Existing System

We studied the existing system related to event management and attendance tracking. In many cases, event organizers still depend on manual methods such as paper-based attendance, simple QR codes without validation, or verbal confirmations to manage participants and track attendance.

We interacted with students and event organizers to understand the current event management process. During the discussion, the following questions were asked:

Q1. How do you manage event registrations and attendance?
Q2. Do you get accurate attendance records?
Q3. Do you face issues with fake or proxy attendance?
Q4. What problems occur in manual attendance systems?
Q5. Is there any proper system with location verification available?
Q6. How do you verify if participants are actually present at the event?
Q7. Do you face difficulty in managing large number of participants?

After discussing with them, we learnt that :-

Ans 1: Most organizers manage registrations and attendance manually or using basic QR systems. These methods are not fully secure or organized.

Ans 2: No, attendance records are not always accurate because there is no proper verification system.

Ans 3: Yes, fake or proxy attendance is a common issue in existing systems.

Ans 4: Manual systems create problems such as data errors, time consumption, and difficulty in managing records.

Ans 5: Some systems are available, but they lack proper location-based verification and security.

Ans 6: Attendance is usually verified manually or through QR codes without confirming the actual presence of participants.

Ans 7: Yes, managing a large number of participants becomes difficult due to lack of proper system and automation.

Q8. Do you face issues in tracking real-time attendance during events?

Q9. Is it difficult to generate reports and analyze event participation data?

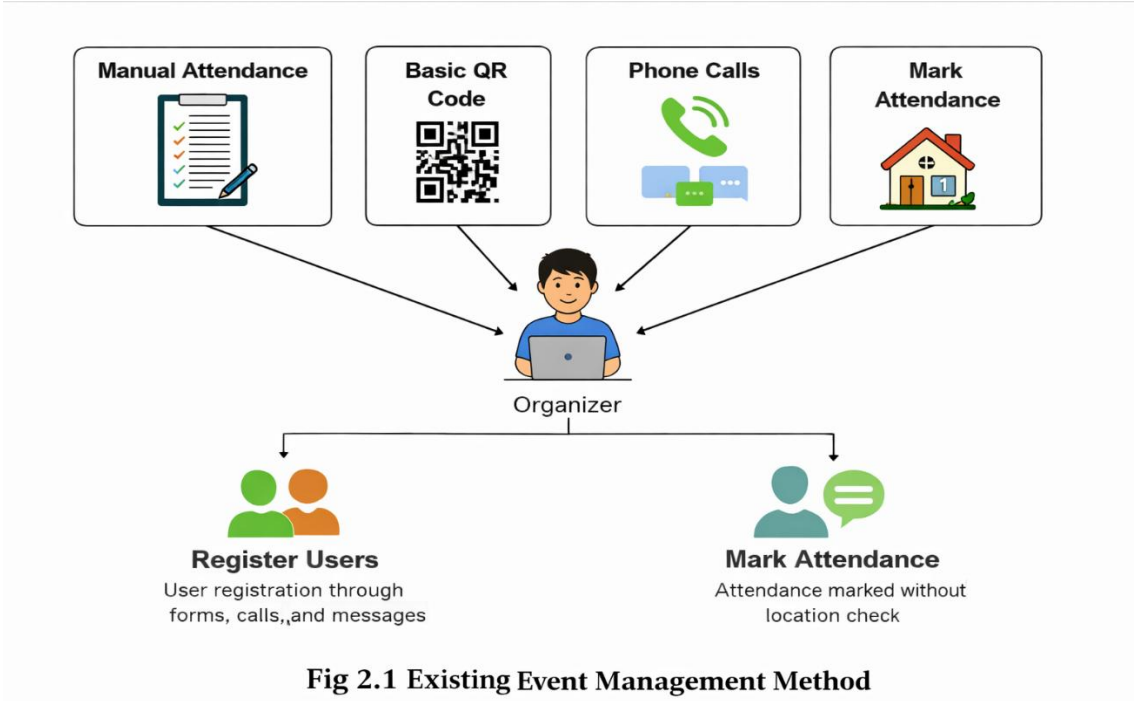

Fig 2.1 Existing Event Management Method

The above figure shows the existing method used for managing events and attendance. In this system, organizers use manual methods, basic QR codes, or communication through calls and messages to handle event registrations and attendance. Participant details are often recorded manually, and attendance is marked without verifying the actual location of the user.

This method is simple but has several limitations. There are chances of fake attendance, inaccurate records, and lack of proper verification. It is also time-consuming and not well organized. Organizers may face difficulty in managing large-scale events due to the absence of a centralized and secure system

.

2.2 Limitations of Existing System

The existing system used for managing events and tracking attendance is mostly manual and unorganized. Event organizers often depend on traditional methods such as paper-based attendance, basic QR codes without validation, and communication through calls or messages. Although these methods are commonly used, they have several limitations that affect efficiency and reliability.

One of the major problems is that organizers do not get accurate and verified attendance data. Participants can mark attendance without actually being present at the event location, which leads to fake or proxy attendance. This creates issues in maintaining reliable records and reduces the credibility of the system.

Another limitation is the absence of a centralized platform. Event-related information such as participant details, attendance records, and event schedules are often scattered across different sources. This makes it difficult for organizers to manage and monitor events effectively.

The system is also time-consuming. Organizers need to manually manage registrations, verify participants, and track attendance, which increases effort and delays the overall process. Handling large numbers of participants becomes even more difficult without automation.

There are also chances of data errors and manipulation. Since there is no proper verification mechanism, attendance data may not always be accurate. This can affect decision-making and reporting.

Communication between organizers and participants is not properly managed. Information is often shared through messages or calls, making it difficult to track and maintain proper records.

Another limitation is the difficulty in filtering and managing participant data. Organizers cannot easily sort or analyze data based on attendance, event participation, or user details, which reduces system efficiency.

Finally, there is no proper admin control in the existing system. There is no centralized authority to monitor, verify, and manage event data, which reduces reliability and creates confusion.

Finally, there is no proper admin control in the existing system. No one manages or verifies

the room listings. This reduces reliability and creates confusion for users.

Summary of Limitations

• Inaccurate and unverified attendance records

• No centralized platform for event management

• Time-consuming manual processes

• Chances of fake or proxy attendance

• Poor communication management

• Difficulty in managing and filtering participant data

• No proper admin verification or control

2.3 Problem Identification

The study of the existing system shows that there are several problems faced by organizers and participants while managing events and attendance. These problems mainly arise due to manual methods and lack of proper verification systems.

Problems of Participant-related Issues

• Participants cannot verify proper event details easily
• Lack of accurate and transparent attendance system
• Possibility of fake attendance affecting genuine users
• Difficulty in tracking event participation history
• Limited access to organized event information

Problems of Organizer-related Issues

• Difficulty in managing large number of participants
• Lack of proper system to track real-time attendance
• No verification mechanism for participant presence
• Time-consuming manual processes
• Difficulty in maintaining accurate records and reports

Problems of Organizer/Management related issues

• Organizers cannot easily manage and track all participants effectively
• Difficulty in handling event registrations for a large number of users
• No proper system to manage event details and attendance records
• Hard to update or modify event information once published
• Communication with multiple participants becomes difficult

Problems of Existing System (Manual System)

• No centralized platform for event management
• Inaccurate or unverified attendance records
• Time-consuming manual processes
• Chances of fake or proxy attendance
• Difficulty in managing and filtering participant data
• No proper admin control or verification

Overall Problems

• Lack of proper digital platform for event management
• Difficulty in managing and tracking attendance accurately
• Unorganized participant and event data
• Inefficient communication between organizers and participants
• Time-consuming and unreliable system

Category	**Problems**
User	Difficulty in event registration, inaccurate attendance, time-consuming process
Organizer	Hard to manage events, participant data, and attendance tracking
Admin	No proper verification, difficulty in monitoring system
System	No centralized database, chances of fake attendance, lack of automation

Table 2.3 Problem Analysis

Fig 2.3 Problem Identification Flow

The above figure shows how the existing event management system works. Users first log into the system and register for events or access event details. The data is stored in the database, but there is no proper verification of participant presence. Attendance is marked without validating the actual location of the user.

2.4 Problem Definition

The existing system for managing events and attendance is mostly manual and unorganized, which creates several challenges for organizers and participants. Due to these limitations, there is a need to develop an efficient and reliable system that can manage event registrations, attendance tracking, and participant verification in a better way.

The main objective of this project is to develop a web-based Event Management System with Geofencing that will help organizers manage events efficiently and ensure accurate attendance tracking. The proposed system aims to solve the problems faced in the existing system by providing a centralized digital platform with secure and location-based verification.

The system will focus on:

a) Maintaining proper records related to event details and participant information

b) Managing event registrations and attendance in an organized manner

c) Providing easy access to event data for users

To achieve this, the proposed system will provide:

a) Detailed categorization of event records:
The system will store event data in a structured format including event name, location, date, time, and participant details. This will help in organizing data properly.

b) Provision to maintain event details:
The system will allow organizers to add, update, and view event details easily. This will reduce manual work and improve data accuracy.

c) Search and filtering system:
The system will provide search options to find events based on requirements such as date, location, and category.

d) Centralized database management:
All records will be stored in a centralized database, making it easy to access data anytime and from anywhere.

e) Admin verification:
The admin will verify event details and registrations before displaying them in the system. This helps reduce errors and improves reliability.

Benefits of Proposed System

- Reduces manual work and effort
- Improves accuracy of event and attendance data
- Saves time for users and organizers
- Provides centralized platform for event management
- Improves communication between organizers and participants

Chapter 3

Scope of the Project

3.1 Scope of the Project

The Event Management System with Geofencing is designed to provide an easy and efficient way for organizers to manage events and track attendance securely. The scope of this project includes developing a web-based platform where users can register for events, organizers can manage events, and attendance can be tracked using geofence-based verification.

The project focuses on providing a centralized platform for managing event details and participant information. Users can register in the system, log in, and view available events based on their preferences. They can also register for events and participate in them easily.

The system also includes an admin module to manage events and user data. The admin verifies event details and ensures proper functioning of the system. This improves transparency and reduces chances of errors or misuse.

Key Functional Areas of the System

The scope of this project is divided into several functional modules which help in efficient management of the system.

1) User Registration and Login

• The system allows users to register using basic details such as name, email, and password.
• Each user will have a unique account in the system.
• Users can securely log in to access system features.
• The system allows users to update and manage their profile details.

2) Event Management Module

• The system allows organizers to create and manage events.
• Users can view event details such as location, date, and time.
• The system helps users explore and participate in different events.

• It reduces manual effort and improves efficiency..

3) Event Registration Module

• Users can register for events by entering their details.
• Details such as name, event selection contact information are stored in the system.
• Registered participants are saved in the database.
• This helps organizers manage participants easily..

4) View Event Details

• The system displays complete event information to users.
• Users can check details such as event location, date, and time.
• This improves decision making for participants.
• It helps users explore multiple events effectively.

5) Participate in Event

• Users can participate in events after registration.
• Attendance is marked using QR code and geofencing.
• This ensures only valid participants are counted.
• It improves system security and reliability.

6) Admin Verification and Management

- Admin verifies room posts before displaying them.
- Admin can manage users and room data.
- Invalid or fake posts can be removed.
- This improves reliability of the system.

7) Data Storage and Security

• All data is stored in a centralized database.
• It ensures safe and organized data management.
• Secure login protects user information.
• Data can be accessed easily when required.

Benefits of the Proposed System

- Reduces manual work and improves efficiency
- Improves accuracy of event and attendance data
- Saves time for users and organizers
- Provides centralized platform for event management
- Improves communication between users
- Reduces chances of fake attendance

Future Scope of the Project

The system can be further enhanced with additional features such as:

- Mobile application for easy access
- Chat system for communication between users and organizers
- Map integration for accurate geofence location tracking
- Advanced search and filtering options
- Cloud-based storage for better accessibility

Chapter 4

Methodology

4.1.1 Requirements Gathering and Analysis

For the development of the Event Management System with Geofencing, we have adopted the Waterfall Model as the software development approach. The Waterfall Model is a linear and sequential process where each phase is completed before moving to the next phase. This model is suitable for this project because the requirements are clearly defined and structured.

The Software Development Life Cycle (SDLC) followed in this project includes the following phases:

1. Requirements Gathering and Analysis
2. System Design
3. Testing
4. Implementation
5. Deployment

Phase 1: Requirements Gathering and Analysis

In this phase, we focused on understanding the needs of users and organizers involved in event management. The main objective was to identify the problems in the existing system and define the requirements of the proposed system.

We collected information through:

• Discussion with students and event organizers
• Observation of existing event management methods
• Analysis of online event platforms

This helped us to design a system that solves real-world problems effectively.

4.1.2 User Requirements

The system should fulfill the following requirements for users:

- Users should be able to register and log in to the system
- Users should be able to view available events
- Users should be able to check event details
- Users should be able to register for events
- Users should be able to mark attendance using geofencing
- The system should be simple and user-friendly

4.1.3 Admin Requirements

- Admin should be able to verify event details
- Admin should manage users and event data
- Admin should remove invalid or fake entries
- Admin should monitor system activities
- Admin should manage database records

4.1.4 Organizer Requirements

- Organizers should be able to create and manage events
- Organizers should be able to view registered participants
- Organizers should track attendance of participants
- Organizers should manage event details
- Organizers should monitor event performance

4.1.5 Event Search Requirements

- The system should allow users to search events by location
- Users should view event details easily
- It should display event information such as date, time, and location
- Users should explore and compare multiple events
- It should provide easy navigation

4.1.6 Record Management Requirements

- The system should store all event and participant data in a structured format
- It should allow adding and updating event records
- It should maintain user and attendance data
- It should provide easy access to stored data
- It should store data securely in the database

4.1.7 System Requirements

Software Requirements

Operating System: Windows OS

Platform: Visual Studio Code

Languages: HTML, CSS, JavaScript

Backend: Node.js

Database: MongoDB

Hardware Requirements

Processor: Intel i3 or higher

RAM: Minimum 4 GB

Hard Disk: 500 GB or more

Phase 2: System Design

In this phase, the system architecture, database structure, and user interface were designed.

- Database design includes tables for users, babies, and vaccination records
- User interface is designed for easy navigation
- System modules are defined clearly

Phase 3: Implementation

In this phase, the system is developed using:

- Frontend: HTML, CSS, JavaScript, React
- Backend: Node.js
- Database: MongoDB

Phase 4: Testing

Testing is done to ensure the system works correctly.

- Unit Testing – testing individual modules
- Integration Testing – testing combined modules
- System Testing – testing the complete system

Errors and bugs are identified and fixed.

Phase 5: Deployment

- The system is deployed on a server
- Users can access the system through a web browser
- System becomes ready for real-time use

Phase 6: Maintenance

- Regular updates are performed
- Bugs are fixed if found
- System performance is improved over time

Chapter 5

Details of Design, Working and Processes

Phase 2: System Design

In the System Design phase of the Event Management System with Geofencing, the main objective was to convert the gathered requirements into a structured and well-organized system. This phase plays a crucial role in defining how the system will function and how different components will interact with each other.

The system design includes:

- System Architecture
- Database Design
- Module Design
- User Interface Design

The design ensures that the system is efficient, user-friendly, and capable of handling all required operations smoothly.

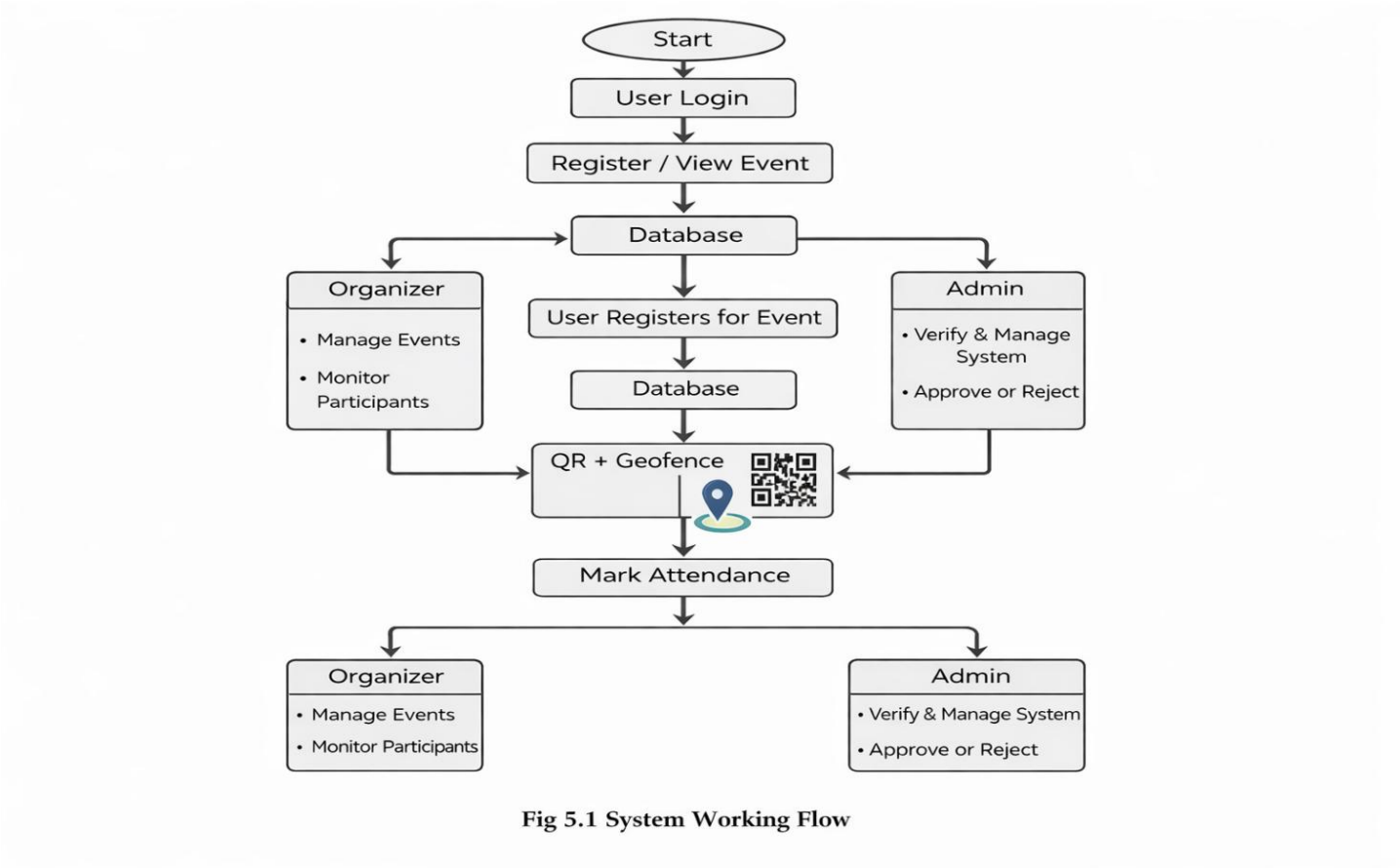

Fig 5.1: System Workflow Diagram

The above figure represents the overall working of the Event Management System with Geofencing, which involves three main users: User, Organizer, and Admin. The system is designed to ensure smooth communication and efficient management of event registration and attendance processes.

Initially, the process starts with the user. The user registers into the system, logs in, and views available events. Based on preferences such as location, date, and type of event, the user can view event details and select a suitable event.

Once the event is selected, the user can register and participate in the event. Attendance is marked using QR code along with geofencing to ensure that the participant is physically present at the event location.

After registration, the details are stored in the system database. The organizer can manage event details and track participants. The admin verifies event data and ensures proper functioning of the system. Once verified, the event becomes visible to users.

The admin plays a vital role in managing the entire system. The admin is responsible for managing users, verifying event details, removing invalid data, and monitoring system activities. The admin also manages the database and ensures smooth functioning of the system.

Step-wise Working of System

1. User registers and logs into the system
2. User views available events
3. System displays event list with details
4. User selects an event and views full information
5. User registers for the event
6. User marks attendance using QR + Geofence
7. Event and attendance data is stored in database
8. Organizer manages event and participant details
9. Admin monitors system activities
10. Admin verifies event data
11. Admin approves or manages system operations

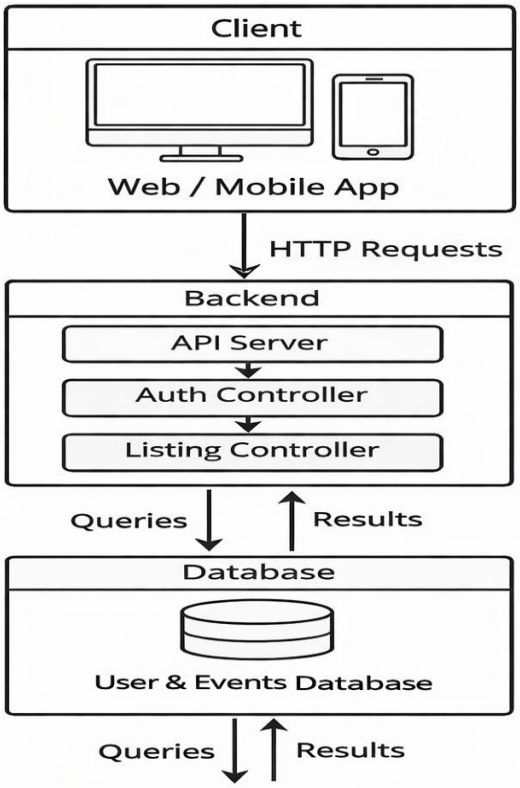

Fig 5.2 System Architecture Diagram

Fig 5.2 System Architecture Diagram

The System Architecture Diagram represents the overall structure of the Event Management System with Geofencing. It shows how different components interact with the system and how the system is organized in a structured manner. The architecture consists of three main layers: Client Layer, Backend Layer, and Database Layer.

The system starts with the client side where users access the application using a web browser or mobile device. The client sends HTTP requests to the backend server. The backend processes the request using API controllers such as authentication controller and event controller. After processing, the server communicates with the database and returns the results to the user.

The database stores all important data such as user details, event information, and attendance records. The architecture ensures proper separation of functionalities, secure data handling, and smooth interaction between users and the system.

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Modules in System Architecture

1.User Module

- • Register and login into the system
  • Create and update profile
  • Search available events
  • View event details
  • Register for events
  • Mark attendance using QR + Geofence

2. Organizer Module

• Create and manage events
• Update event details
• Monitor participants
• View attendance records
• Manage event activities

3. Participant Module

• Search available events
• View event details
• Register for events
• Mark attendance using geofencing
• View participation history

4. Admin Module

• Manage users and profiles
• Verify event details
• Remove fake or invalid entries
• Monitor system activities
• Manage database records

Key Features of Architecture

• Role-based access control system
• Centralized database management
• Secure and structured data flow
• Real-time data processing

Fig. 5.3 Use Case Diagram

Actors

- User: Represents a participant who registers for events and marks attendance
- Organizer: User who creates and manages events
- Participant: User who attends events
- Admin: Manages the entire system

Use Cases

Authentication

- Register: User creates account in system
- Login: User logs into system
- Profile Creation: User adds personal details
- Verification: System validates user credentials

User Activities

- Search Events: User searches available events
- View Events: User views event details
- Register Event: User registers for selected event
- Mark Attendance: User marks attendance using QR + Geofence
- Update Profile: User updates personal details

Organizer Activities

- Create Event: Organizer posts event details
- Manage Events: Organizer updates event information
- View Participants: Organizer sees registered users
- Track Attendance: Organizer monitors attendance
- Contact Users: Organizer communicates with participants

Participant Activities

- Search Events: Participant searches events
- View Details: Participant checks event details

- Register for Event: Participant enrolls in event
- Mark Attendance: Attendance marked using geofencing
- Update Profile: Participant updates personal details

Admin Activities

- Manage Users: Admin controls user accounts
- Verify Events: Admin approves event details
- Remove Data: Admin deletes invalid entries
- Monitor System: Admin monitors system activities
- Generate Reports: Admin creates reportsx`

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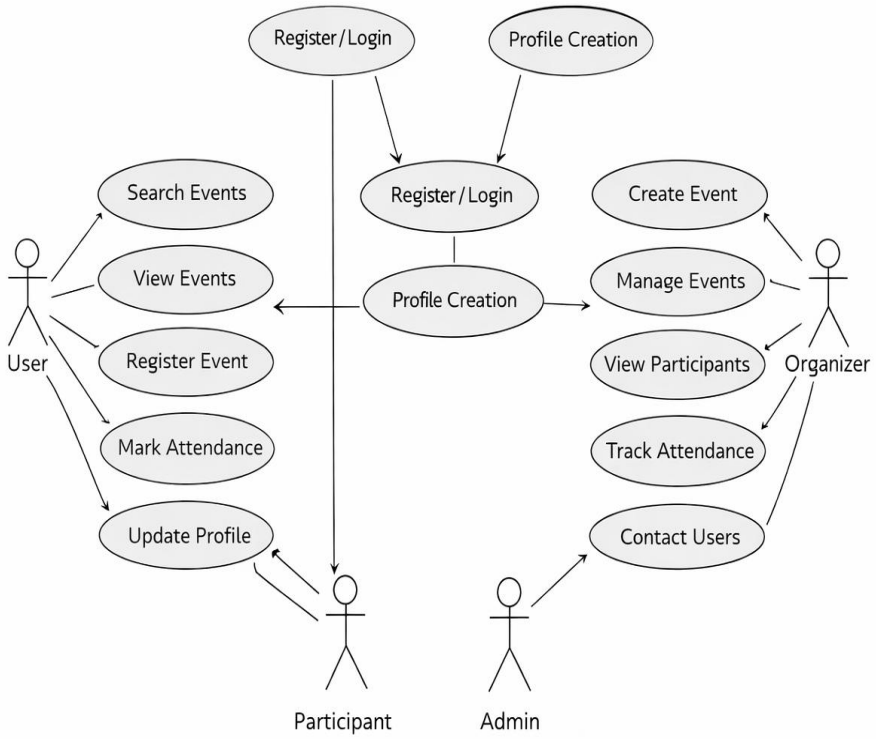

Fig 5.3 Use Case Diagram

Fig. 5.4 ER Diagram

The above use case diagram represents the interaction between different actors and the Event Management System with Geofencing. It shows how users, organizers, and admin interact with the system to perform various operations such as registration, event management, and attendance tracking. The diagram highlights the role-based functionalities and ensures smooth workflow within the system.

Chapter 6

Results and Applications

The Event Management System with Geofencing has been successfully developed and implemented as a web-based application to manage events and track attendance efficiently. The system provides a digital platform for users, organizers, and administrators to interact and manage event-related activities in a structured and organized manner.

The application is developed using HTML, CSS, JavaScript, React, and Node.js, with a backend database managed using MongoDB. The system ensures easy accessibility, secure data handling, and efficient management of event and attendance information.

1. User Module Results

The user module allows users to search and find rooms easily.

- Users can register and log in into the system
- Users can view available events
- Users can check event details such as location, date, and time
- Users can register for events
- Users can mark attendance using QR code and geofencing

This module helps users find suitable accommodation quickly and reduces manual searching effort.

2. Organizer Module Results

The room owner module helps users post room availability.

- Organizers can register and log in
- Create and manage event details
- Update or delete events
- View registered participants
- Track attendance of participants

This module helps owners find tenants or flatmates easily.

3. Participant Module Results

The flatmate module helps users find shared accommodation.

- Users can create profile and set preferences
- Search available events
- Register and participate in events
- Mark attendance using QR code and geofencing
- Update profile and preferences anytime

This improves communication and helps users find compatible flatmates.

4. Admin Module Results

The admin module provides complete control over the system.

- Admin manages users and profiles
- Admin verifies event details
- Admin removes fake or invalid data
- Admin monitors system activities
- Admin manages database records

Admin ensures proper functioning, reliability, and security of the system.

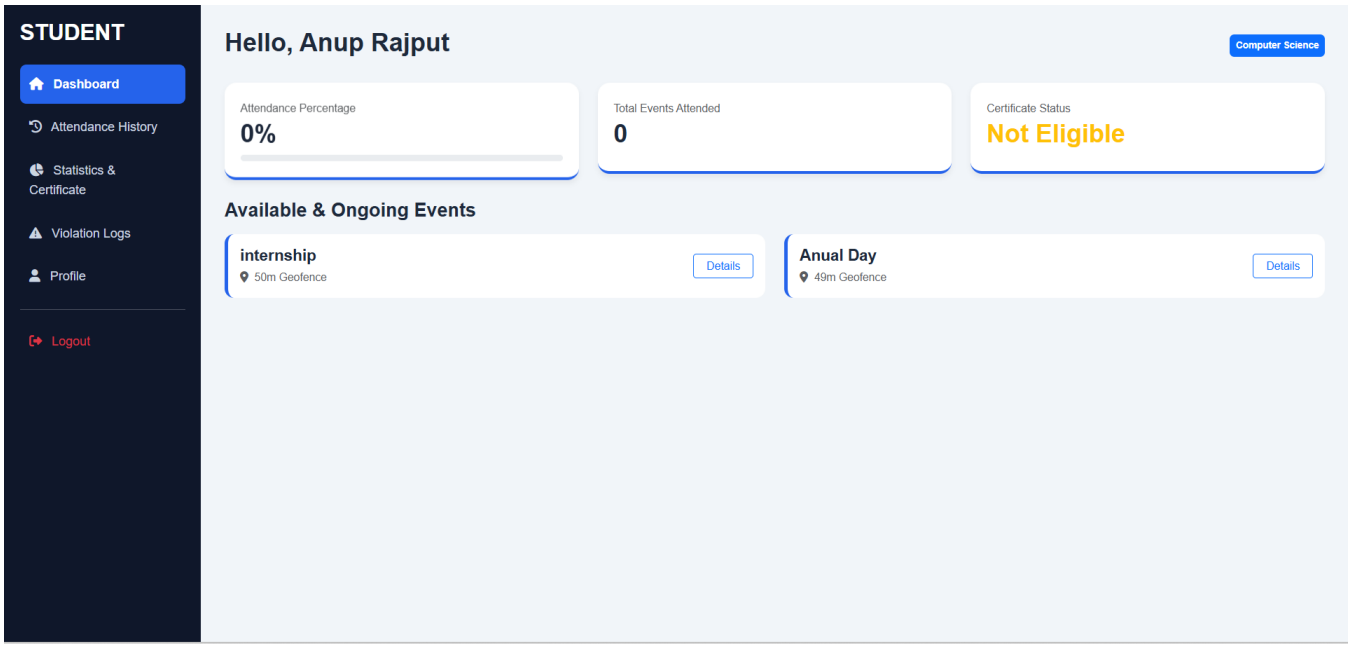

Fig 6.1: Dashboard Interface

The above diagram represents the Home Dashboard of the Event Management System with Geofencing. It provides a user-friendly interface for users to explore events easily. The dashboard displays important options such as search functionality, registration, and login for accessing system features.

At the top of the dashboard, a navigation bar is available which includes the project title and options like Register and Login. These options help new users create an account and existing users access their profiles. The simple navigation improves usability and allows users to move easily within the system.

The main section of the dashboard contains a search bar where users can search for events. Users can enter keywords such as location, event type, or preferences to find suitable events. Below the search bar, popular locations or categories are displayed, which helps users quickly explore available options.

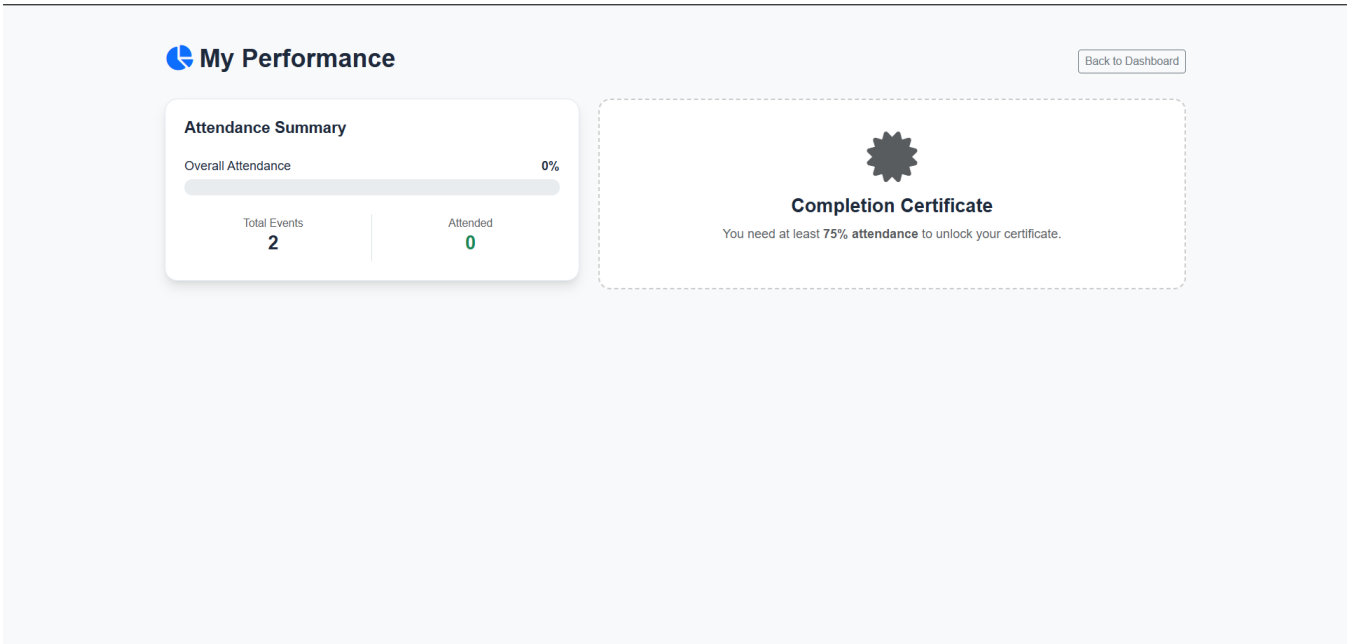

Fig 6.2 Student Performance Page and Attendance.

The above diagram represents the My Performance and Attendance interface of the Event Management System with Geofencing. This interface allows users to view their overall participation and attendance summary for different events. It provides a clear overview of total events attended, attendance percentage, and eligibility for certificates.

The interface is designed to be simple and user-friendly, displaying important details such as attendance statistics and completion status. Users can easily track their progress and performance in events. The system also helps in identifying whether the user has met the required attendance criteria for certification. This feature improves transparency and encourages active participation in events.

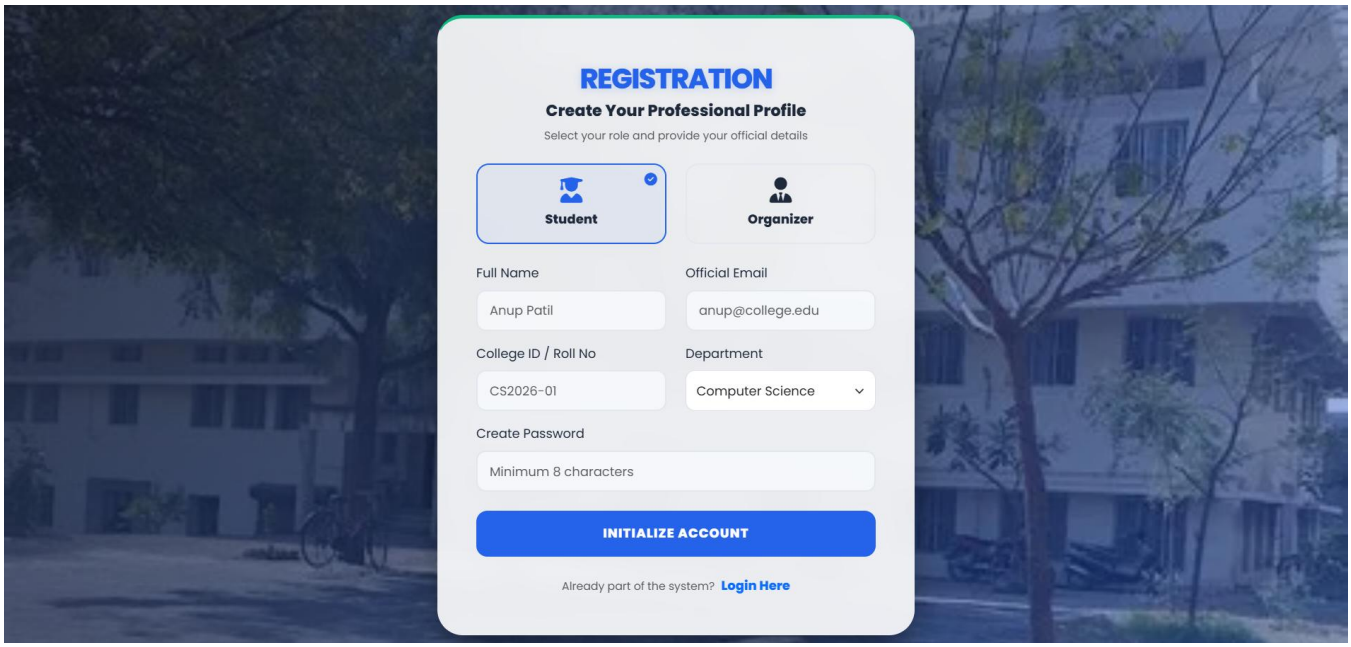

Fig 6.3: User register Page

The above diagram represents the User Registration interface of the Event Management System with Geofencing. This interface allows new users to create an account by entering their basic details such as name, email, role (student/organizer), and password.

The interface is designed to be user-friendly and ensures that users can quickly register into the system. It provides a clean and structured form layout for easy data entry. After filling in the required details, users can initialize their account and access system features. This feature helps in managing user information and enables secure access to the system functionalities such as event participation and attendance tracking.

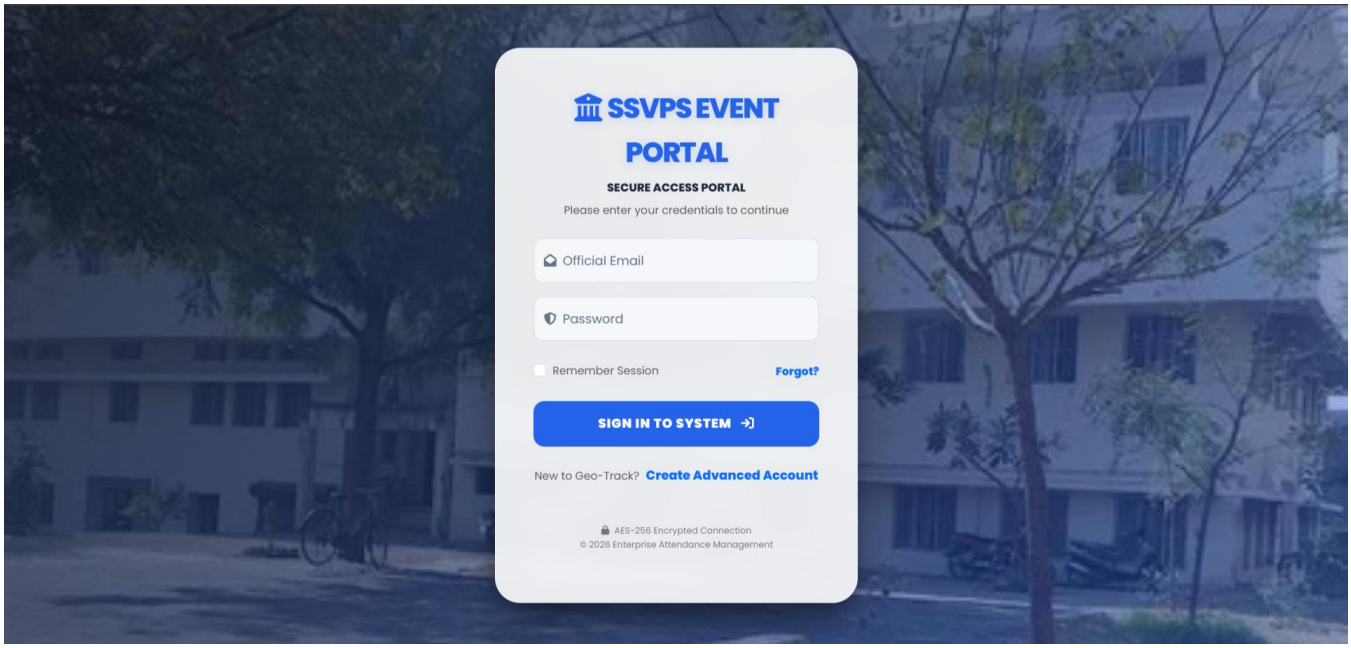

Fig 6.4: User Login Page

The above diagram represents the User Login interface of the Event Management System with Geofencing. This interface allows registered users to securely access the system by entering their login credentials such as email and password.

The interface is designed to be simple and user-friendly, providing input fields for user credentials along with options like password recovery and account creation. It ensures secure authentication and allows users to access system features such as event registration, participation, and attendance tracking.

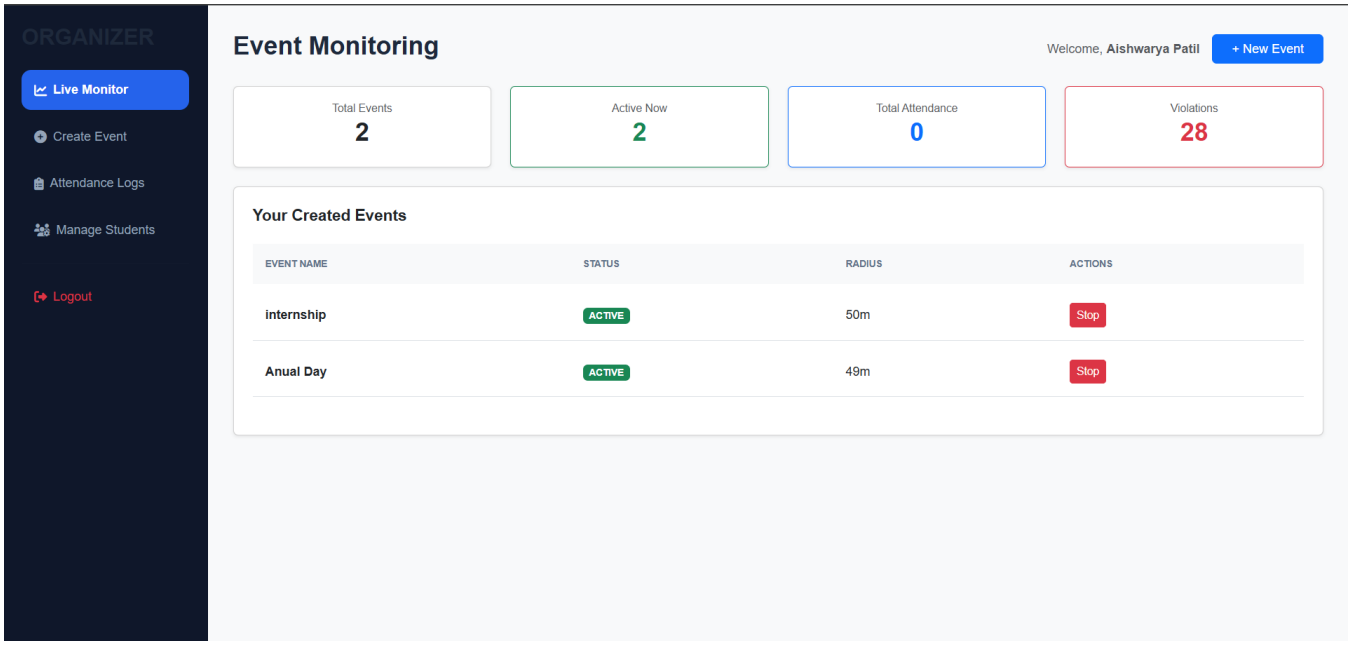

Fig 6.5: Organizer Home Page

The above diagram represents the Organizer Dashboard interface of the Event Management System with Geofencing. This interface provides organizers with a complete overview of their events and activities.

The dashboard displays important information such as total events, active events, attendance count, and participant details. It also provides options to create new events, manage existing events, and monitor attendance in real-time. The interface is designed to be clean and efficient, helping organizers manage events effectively and track performance easily.

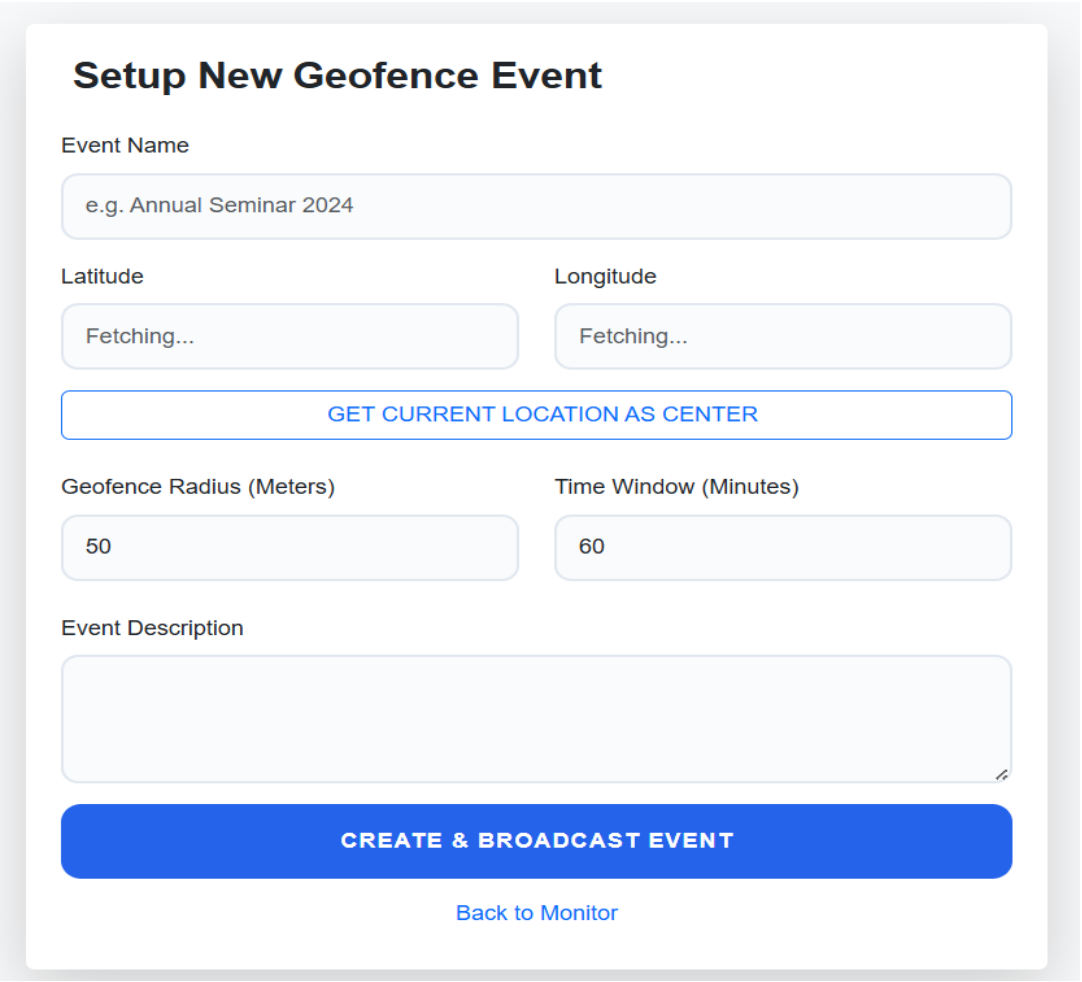

Fig 6.6: Event Registration Portal

The above diagram represents the Geofence Event Creation interface of the Event Management System with Geofencing. This interface allows organizers to set up new events by entering essential details such as event name, location coordinates (latitude and longitude), geofence radius, and time duration.

The system provides an option to automatically fetch the current location, which simplifies the process of defining the event center. Organizers can also add a description of the event to provide additional information. The interface is designed in a clean and structured manner, making it easy to create and broadcast events efficiently. This feature ensures accurate event setup and supports secure geofence-based attendance tracking.

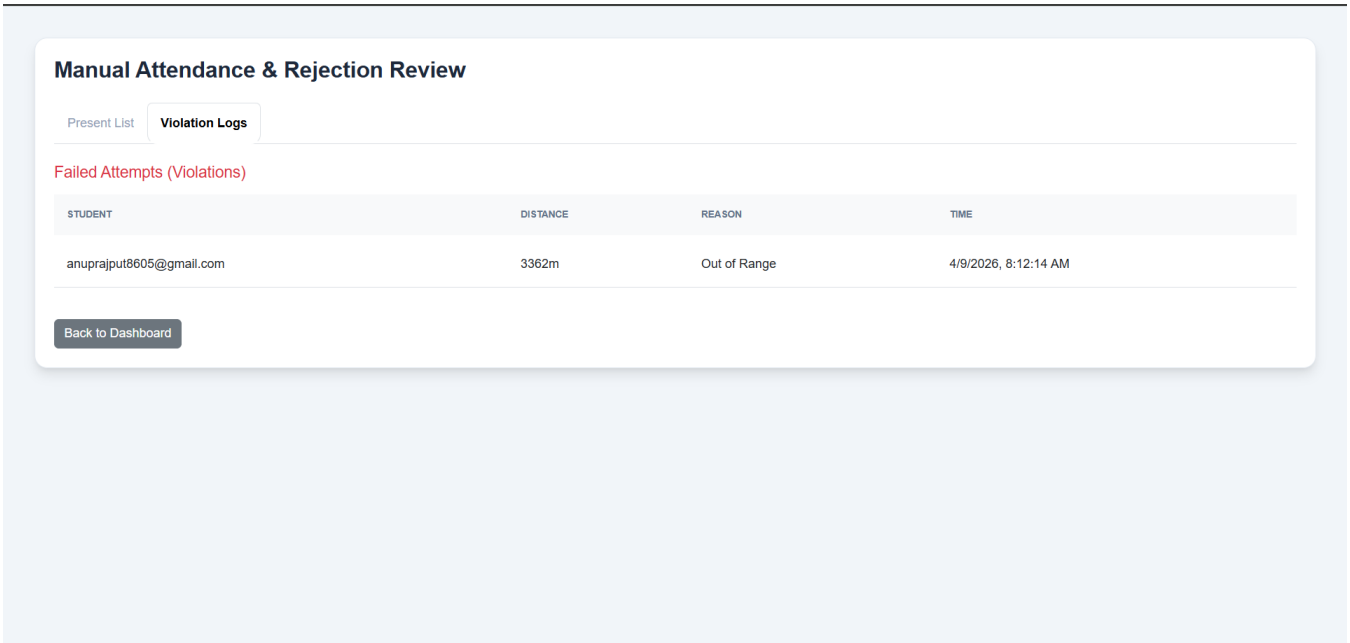

Fig 6.7: Violation logs

The above diagram represents the Violation Logs interface of the Event Management System with Geofencing. This interface displays records of failed or invalid attendance attempts where users try to mark attendance outside the allowed geofence range.

It includes details such as user identity, distance from the event location, reason for rejection (such as "Out of Range"), and time of attempt. This helps organizers and administrators monitor suspicious activities and maintain system integrity. The interface is organized in a tabular format, making it easy to analyze and review violation data effectively.

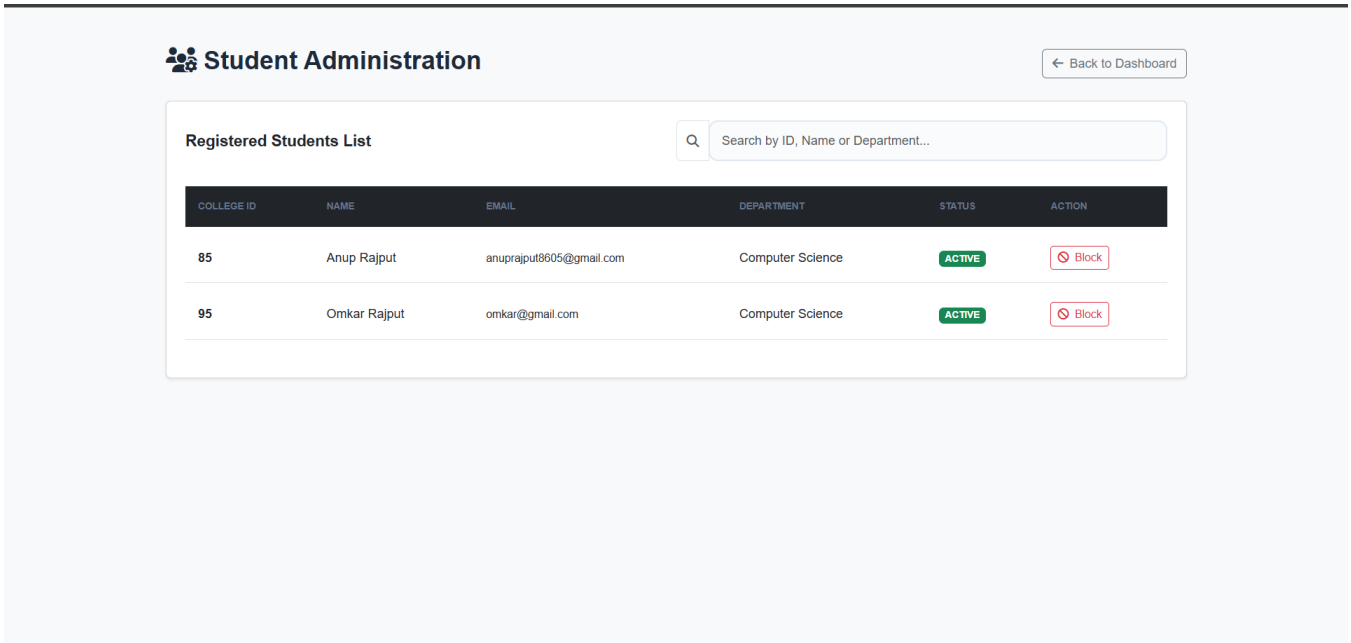

Fig 6.8: Student Administration

The above diagram represents the Student Administration interface of the Event Management System with Geofencing. This interface allows the admin to manage registered users within the system.

It displays a list of users along with details such as user ID, name, email, department, and status. The admin can perform actions such as activating or blocking users based on system requirements. The interface also includes a search option to quickly find specific users. This feature helps in efficient user management and ensures proper control over system access.

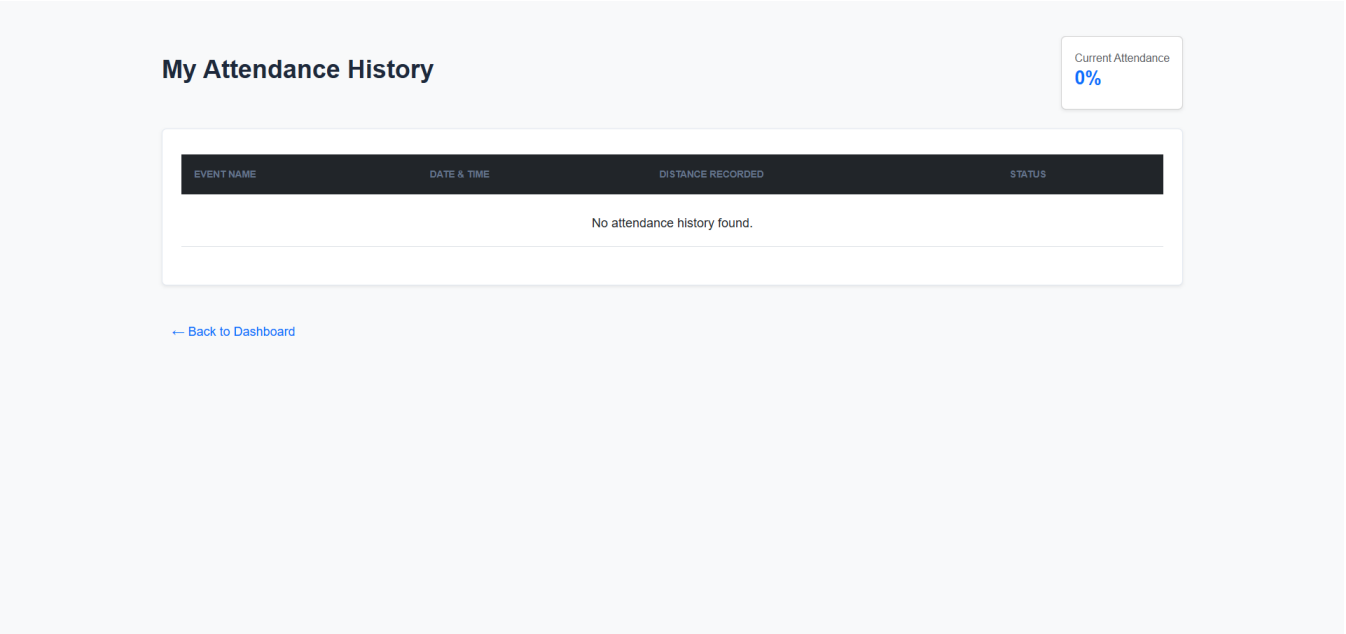

Fig 6.9: Attendance History

The above diagram represents the Attendance History interface of the Event Management System with Geofencing. This interface allows users to view their past attendance records for different events.

It displays details such as event name, date and time, attendance status, and distance records. Users can track their participation and verify whether their attendance was successfully recorded. The interface is simple and organized, providing clear insights into user activity and attendance performance.

Chapter 7

Testing and Defect Report

1. Introduction

The Test Plan for the Event Management System with Geofencing outlines the overall strategy, scope, objectives, and methodology for testing the system. Testing is a crucial phase in the software development lifecycle, as it ensures that the system functions correctly, meets user requirements, and is free from critical defects. This test plan provides a structured approach to verify and validate the system functionalities.

2. Objectives of Testing

The main objectives of testing the system are as follows:

- To verify that all modules of the system are working as expected
- To identify defects, errors, and inconsistencies in the system
- To ensure that the system meets functional and user requirements
- To validate data accuracy and system performance
- To improve system reliability, usability, and efficiency

3. Scope of Testing

The testing process covers all major modules of the Baby Vaccination Management System, including:

- User Registration and Login Module
- Event Management Module
- Attendance Tracking (QR + Geofence) Module
- Admin Management Module
- Dashboard and Reporting Module

4. Testing Methodology / Approach

The following testing approach was used:

- Manual Testing: All functionalities were tested manually using predefined test cases
- Functional Testing: Each feature was tested to ensure correct operation
- Integration Testing: Interaction between modules was verified

• System Testing: Complete system testing was performed to ensure end-to-end functionality

This approach helped in identifying defects at different levels of the system.

5. Test Environment

The testing was performed in the following environment:

• Operating System: Windows 10/11
• Frontend Technologies: HTML, CSS, JavaScript
• Backend Technology: Node.js
• Database: JSON

This environment ensured proper execution and validation of all system functionalities.

6. Test Data

Various types of test data were used to validate system behavior:

• Valid login credentials
• Invalid username and password combinations
• Sample user and event details
• Event data (location, time, radius settings)
• Attendance records (QR scans and geofence validation)
• Edge cases such as empty fields, incorrect inputs, and duplicate entries

7. Entry Criteria

Testing was initiated based on the following conditions:

• Development of all modules was completed
• Database was properly configured
• Required test cases were prepared
• System was deployed in a testing environment

8. Exit Criteria

Testing was considered complete when:

• All major test cases were executed
• Critical and high-severity defects were resolved
• System performance met expected standards
• The system was ready for deployment

9. Risks and Challenges

During testing, the following risks and challenges were identified:

• Possibility of data validation errors
• Performance issues due to large data handling
• UI inconsistencies on different devices
• Dependency on internet connectivity and server availability

10. Limitations of Testing

• Testing was conducted in a controlled environment
• Real-time user load testing was limited
• Some minor UI issues may still exist
• External integrations were not fully tested

11. Conclusion

The Test Plan provided a systematic and structured approach for testing the Event Management System with Geofencing. It helped in identifying defects, improving system performance, and ensuring that all functionalities work as expected. After successful testing and fixing of identified issues, the system was found to be reliable, efficient, and ready for deployment.

Testing and Validation

Testing is a crucial phase in the development of the Event Management System with Geofencing. It ensures that the system functions correctly, meets user requirements, and is free from errors. Various types of testing were conducted to verify the performance, functionality, and reliability of the system. The main objective of testing is to identify defects and ensure smooth operation of the application.

- **Types of Testing Performed**

1) Unit Testing – Individual modules like login, event registration, and attendance tracking were tested separately
2) Integration Testing – Interaction between modules such as user, organizer, and admin was tested
3) System Testing – Complete system functionality was tested to ensure proper working
4) User Acceptance Testing – System was tested by users for real-time validation

Table no 6.2.1 Test cases

Test Case ID	Module	Description	Input	Expected Output	Actual Output	Status
TC01	Login	Valid login credentials	email & password	Login successful	Login successful	Pass
TC02	Login	Invalid password	wrong password	Error message	Error displayed	Pass
TC03	Registration	Empty fields	blank input	Validation error	Error shown	Pass
TC04	User Module	Add user details	valid data	Data saved	Data saved	Pass
TC05	User Module	Missing email	empty email	Error message	Error shown	Pass
TC06	Event Creation	Create event	valid data	Event created	Created	Pass
TC07	Event Creation	Missing event name	empty field	Not allowed	Error shown	Pass
TC08	Event Search	Search events	valid location	Event list displayed	Displayed	Pass
TC09	Event Creation	Missing location	empty field	Error	Accepted	Fail
TC10	Dashboard	Load dashboard	user login	Data displayed	Loaded	Pass
TC11	Attendance	Mark attendance	QR + valid location	Attendance marked	Marked	Pass
TC12	Admin	Verify event	valid event	Approved	Approved	Pass
TC13	Profile	Update profile	valid input	Updated	Updated	Pass
TC14	Security	Invalid QR/spoof	malicious input	Blocked	Blocked	Pass
TC15	Event Creation	Duplicate event	same data	Not allowed	Allowed	Fail

Failed Test Cases and Defects

During the testing phase, several defects were identified in the Event Management System with Geofencing which affected functionality and user experience. These issues were carefully analyzed to improve system performance and reliability.

The event creation module allowed submission of events with empty or incomplete fields in certain scenarios, indicating insufficient input validation. This could lead to incorrect or incomplete event information being stored in the system.

The system also permitted duplicate event entries for the same details, which could create confusion for users. This highlighted the need for proper duplicate validation mechanisms.

There was a minor delay observed while loading the dashboard, especially when handling larger datasets, affecting system responsiveness. This required optimization of database queries and frontend rendering.

These defects were further reviewed and necessary corrective measures were taken to enhance the overall system performance and user experience.

Table no 6.2.2 Defect Report

Defect ID	Module	Description	Severity	Status
D01	Event Creation	System allowed submission of events with empty or missing fields due to improper validation	Medium	Fixed
D02	Event Creation	System allowed duplicate event entries for same details causing confusion	High	Fixed
D03	Dashboard	Dashboard experienced slight delay while loading data, affecting user experience	Low	Fixed
D04	UI	User interface alignment issues observed on different screen sizes, especially in mobile view	Low	Pending

Testing Summary

The Event Management System with Geofencing was thoroughly tested using multiple test cases that covered all major functionalities of the application, including login, registration, event creation, event search, attendance tracking, profile management, and admin verification. The primary objective of testing was to ensure that the system operates correctly, efficiently, and according to user requirements.

During the testing process, most of the test cases were executed successfully, indicating that the core functionalities of the system are working as expected. The system was able to handle user inputs properly, perform required operations, and display accurate results in different modules.

However, a few defects were identified during testing. These issues included validation errors, duplicate event entries, minor performance delays, and user interface inconsistencies. Each defect was carefully analyzed based on its severity and impact on the system. Appropriate corrective measures were taken to resolve critical and medium-level issues, while minor issues were noted for further improvement.

After implementing the necessary fixes, the system showed significant improvement in terms of performance, reliability, and usability. The final testing results indicate that the system is stable, user-friendly, and capable of handling real-time event operations effectively.

Overall, the testing phase played a crucial role in enhancing the quality of the system and ensuring that it meets the desired functional and performance standards.

Chapter 8

Conclusion and Future Scope

Conclusion

In conclusion, the Event Management System with Geofencing provides an efficient and user-friendly platform for managing events and tracking attendance securely. The system allows users to register for events, organizers to create and manage events, and admins to monitor activities in an organized manner. It simplifies traditional event management processes and reduces manual effort.

The system provides a centralized platform where all event details and attendance records are stored and managed properly. Users can easily view event details such as location, time, and description before participating. The geofencing feature ensures that attendance is marked only when users are within the specified location, improving accuracy and preventing fake attendance.

This system reduces dependency on manual verification and saves time for organizers and participants. It also helps organizers manage events efficiently and monitor real-time participation. The simple interface and structured database management improve overall system performance. The application ensures secure data handling and smooth communication between users.

Overall, the Event Management System with Geofencing enhances event management processes, improves user experience, and provides a reliable digital solution for secure attendance tracking.

Future Scope

Mobile Application Development

A dedicated mobile application for Android and iOS can be developed to make the system more accessible. Users will be able to register, join events, and mark attendance directly from their smartphones. This will improve usability and allow users to receive instant updates anytime and anywhere.

Multi-Language Support

The system can be enhanced by adding multi-language support so that users from different regions can use the platform easily. Languages such as English, Hindi, and regional languages

can be included, making the system more user-friendly and accessible.

Chat and Messaging System

A real-time chat and notification system can be integrated to allow communication between organizers and participants. Users will receive instant notifications about event updates, reminders, and announcements, improving engagement and communication.

Map and Location Integration

The system can include advanced map integration to show exact event locations. Users will be able to navigate to event venues and view nearby areas easily. This will improve location accuracy and enhance user experience.

Advanced Search and Filter Options

Advanced filtering options such as event type, date, location, and category can be added. Users will be able to quickly find relevant events based on their preferences, improving efficiency and usability.

Cloud-Based Storage and Scalability

The system can be migrated to cloud-based storage for better performance and scalability. Cloud hosting will allow the system to handle a large number of users and events efficiently while improving data security, backup, and availability.

Chapter 9

References and Bibliography

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